LECTURE 9

Review

What is the domain of $\sum_{n=0}^{\infty} x^n$?

By the geometric series test, it converges when $|r| = |x| < 1$. So the domain is $-1 < x < 1$.

For $x \in (-1, 1)$, we get $\sum_{n=0}^{\infty} x^n = \frac{x^0}{1-x} = \frac{1}{1-x}$

<u>The Ratio Test</u>

$$\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| \begin{cases} < 1 & \Rightarrow \text{absolutely convergent} \\ > 1 & \Rightarrow \text{divergent} \\ = 1 & \Rightarrow \text{test fails} \end{cases}$$

Def'n

A series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \cdots$$

is called a <u>power series</u>.

We call the set of all x for which the series converges the <u>interval of convergence</u> (i.o.c.).

We define the <u>radius of convergence</u> (r.o.c.), R, as half of the length of the i.o.c.

Theorem

For a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$:

(1) The power series only converges when $x = a$ So the r.o.c is $R = 0$ and the i.o.c is $x = a$ ($[a, a]$)

(2) It converges for all $x \in \mathbb{R}$. So the r.o.c is $R = \infty$ and the i.o.c is $(-\infty, \infty)$.

(3) There exists a positive finite number, R, the r.o.c such that the power series converges when $|x - a| < R$

power series, interval of convergence, radius of convergence

CONTINUED

To find the r.o.c. we must test the end points of the interval $(-R+a, R+a)$.

Ex Find the r.o.c. and i.o.c. for $\sum_{n=1}^{\infty} \frac{(x-3)^n}{2n^2 \cdot 5^n}$.

To find the r.o.c we must use the Ratio Test.

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(x-3)^{n+1}}{2(n+1)^2 \cdot 5^{(n+1)}} \cdot \frac{2n^2 \; 5^n}{(x-3)^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(x-3)}{5} \cdot \frac{1}{\left(1+\frac{1}{n}\right)^2} \right|$$

$$= \left| \frac{x-3}{5} \right|$$

By the ratio test, the power series test converges when $\left| \frac{x-3}{5} \right| < 1 \Rightarrow |x-3| < 5$

Hence the r.o.c. is $R = 5$.

We have $|x-3| < 5 \Rightarrow -2 < x < 8$.

For $x = -2$, we get $\sum_{n=1}^{\infty} \frac{(-2-3)^n}{2n^2 \cdot 5^n} = \sum_{n=1}^{\infty} \frac{(-1)^n 5^n}{2n^2 \cdot 5^n}$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{2n^2}$$

Let $b_n = \frac{1}{2n^2}$.

1) $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{2n^2} = 0$

2) $b_n = \frac{1}{2n^2} > \frac{1}{2(n+1)^2} = b_{n+1}$

By AST, it converges. So $x = -2$ is in the i.o.c.

CONTINUED

For $x = 8$, we get $\sum_{n=1}^{\infty} \frac{(8-3)^n}{2n^2 \cdot 5^n} = \sum_{n=1}^{\infty} \frac{5^n}{2n^2 \cdot 5^n}$

$$= \sum_{n=1}^{\infty} \frac{1}{2n^2}$$

$$= \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

By the p-series test, it converges.

Thus the i.o.c is $[-2, 8]$.

$$\sum_{n=0}^{\infty} \frac{(x+1)^n}{3^{n+1}}$$

We rewrite as $\sum_{n=0}^{\infty} \frac{1}{3} \left(\frac{x+1}{3} \right)^n$ By GST,

it converges when $\left| \frac{x+1}{3} \right| < 1 \Rightarrow |x+1| < 3$

Hence the r.o.c. is $R = 3$.

By GST, the power series diverges at $x = -4$ and $x = 2$. So the i.o.c. is $(-4, 2)$.